
Joint Learning of Dark Image Enhancement and Optical Flow in the Event-Based Domain

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Abstract

Event cameras provide asynchronous brightness-change measurements with micro-second latency and high dynamic range, making them ideal for extremely dark or high-speed scenes. Existing work tackles low-light enhancement and event-based optical-flow estimation separately, missing their natural synergy. We propose a unified framework that *jointly* enhances low-light RGB frames and predicts dense optical flow. Our architecture extends a state-of-the-art low-light enhancement network by inserting a *Swin-T7 transformer* backbone that acts as a event feature encoder *shared* by both the enhancement and optical-flow branches.

1. Introduction

An event camera is a kind of sensor that records signals when the individual pixels exhibit illumination changes. An event camera reports signals, called events, asynchronously when the log intensity change exceeds a predetermined threshold τ (Gallego et al., 2019). Each event encodes the position, timestamp, and polarity of brightness change, resulting in streams that capture motion and contrast with microsecond precision with higher frame rate, higher dynamic range and lower power consumption. Event cameras are evident in many computer vision tasks, e.g., deblurring (Jiang et al., 2020), optical flow estimation (Liu et al., 2019), high-speed object tracking (Gallego et al., 2019), and SLAM in challenging environments.

Capturing scenes under extreme low-light is one of challenging task: dark frames exhibit low contrast, heavy noise and colour shifts, thwarting high-level vision algorithms. Conventional frame-based low-light enhancement methods struggles below a certain illumination due to amplified noise and color shifts (Liang et al., 2024). Event cameras alleviate these issues by asynchronously reporting per-pixel brightness changes (*events*) with high temporal resolution and no motion blur. However, pure event reconstructions tend to miss fine spatial detail without explicit priors (Liu et al., 2023). Recent work bridges this gap via cross-modal fusion: Jiang et al. (Jiang et al., 2024) show that residual fusion modules can seamlessly blend event streams with

image content, and EvLowLight (Liu et al., 2023) employs a transform-based fusion to align multi-resolution cues for more coherent outputs.

Prior methods treat low-light enhancement and event-based optical-flow estimation as disjoint problems. We argue for a *multi-task* approach: motion cues can guide de-noise/de-blur operations, while an enhanced image offers structure that regularises optical flow. Motivated by recent advances in dense event camera pre-training (Yang et al., 2024) and spatially-guided temporal aggregation (Zhou et al., 2025), we adopt a multi-task strategy. Specifically, we leverage optical flow supervision to inform enhancement networks of motion boundaries, while enhanced image features provide structural priors to regularize flow estimation under low-light conditions.

We adopt the transformer backbone of ECDDP (Yang et al., 2024) and integrate it into the EvLight low-light enhancement pipeline (Zhou et al., 2025) as an *event feature encoder*. We introduce some **bi-directional cross-task consistency losses** that enforces coherence between tasks by aligning strong image gradients with peaks in optical-flow magnitude.

2. Related Work

Low Light Image Enhancement. Low-light imaging in under extreme lighting conditions is a challenging task for both frame-based and event-based models. Conventional enhancement methods degrade significantly when illumination falls below certain threshold, while event-driven approaches lack spatial priors to reconstruct natural-looking images. Jiang et al. (Jiang et al., 2024) utilize a residual fusion module to blend event and image for low light enhancement. Liu et al. (Liu et al., 2023) propose a fusion transform

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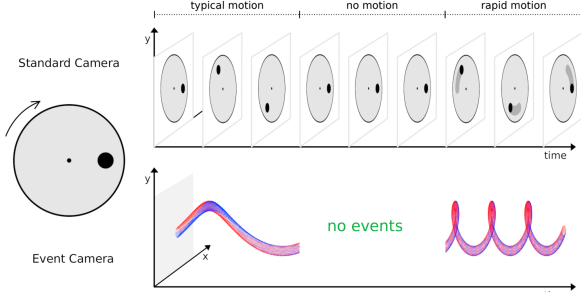


Figure 1. A demonstration of the difference between event and traditional RGB cameras. The event camera only records the differences in brightness between consecutive frames allowing for much faster sampling speeds, compared with a traditional camera that only samples at sparse intervals of time.

module to fuse event features with image features. EvLowLight (Liang et al., 2023) establishes temporal coherence by jointly estimating motion from both events and intensity frames while ensuring the spatial coherence between events and frames with different spatial resolutions. EvLight (Liang et al., 2024) does not directly fuse features extracted from events and images like EvLowLight, instead it considers the discrepancy of the noise at the different regions in events and images, providing SDE dataset containing image+event paired sequences in different environmental conditions. Pre-training techniques tailored for dense predictions on event data have shown promise to mitigate this gap by learning robust representations through self-supervision and cross-modal guidance (Yang et al., 2024).

Optical Flow Estimation. Optical flow estimation based on event streams is a challenging task due to the unique structure of visual information event streams encode. Zhu et al. utilizes the grayscale intensity images along with event streams as the self-supervision signals to train the network (Zhu et al., 2019). Li et al. proposes a lightweight pyramid network with attention mechanism to learn optical flow from events data (Li et al., 2021). Ding et al. proposed STE-FlowNet (Ding et al., 2022), a spatio-temporal recurrent encoding-decoding neural network architecture for event-based optical flow estimation, end-to-end trained in a self-supervised manner. All these approaches use the U-Net architecture to directly estimate and evaluate sparse optical flow from event representations. Gehrig et al. (Gehrig et al., 2021b) proposed the first optical flow approach designed for dense optical flow estimation by introducing cost volumes and recurrency to incorporate temporal priors. Spatially-guided temporal aggregation (SGTA) approaches like that of Zhou et al. (Zhou et al., 2025) reveal that injecting dense RGB features into event pipelines via attention yields more stable optical flow in low-texture regions. Yet, these methods still decouple enhancement from motion, leaving the potential of joint supervision largely unexplored.

Event Camera Pre-training. ECDDP (Yang et al., 2024) proposes the first pre-training framework tailored for dense prediction tasks based on event data without paired RGB images and pre-trained RGB networks. Zhou et al. (Zhou et al., 2025) proposes an approach that uses a spatially dense modality to guide the aggregation of the temporally dense event modality, achieving effective cross-modal fusion and outperforming ECDDP for the task of dense optical flow.

3. The Approach

Our network tackles *low-light enhancement* and *event-based optical flow* within a single, shared backbone. We start from the self-supervised ECDDP student backbone of Yang et al. (Yang et al., 2024), a Swin-T7 encoder that projects raw events into patchwise features. These features are fused with the RGB branch of **EvLight**, the state-of-the-art low-light image enhancement network, yielding a two-stream encoder that is common to both tasks.

- Spatial cues from the RGB image guide a *temporal-aggregation* block, concentrating high-frequency event information along trustworthy image structures.
- Enriched event features are injected into the image especially where RGB features are degraded

From the shared representation we attach two heads: (i) an enhancement decoder that produces the restored image I_{en} , and (ii) a pyramid flow head that outputs the dense motion field V . The two branches are coupled by **bidirectional consistency losses**—photometric, flow-warping, and temporal terms—that align strong image gradients with peaks in flow magnitude and enforce forward-backward coherence.

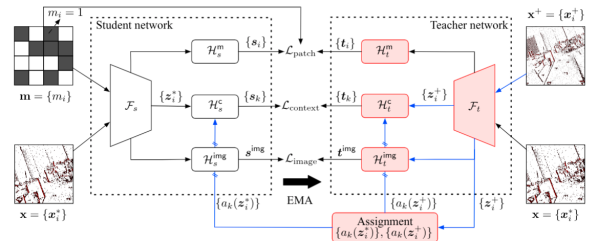


Figure 2. The ECDDP pre-training framework

Problem formulation. Given a low-light RGB frame $I \in \mathbb{R}^{H \times W \times 3}$ and a stream of events $\mathcal{E} = \{e_k\}_{k=1}^N$, where $e_k = (x_k, y_k, t_k, p_k)$ with polarity $p_k \in \{-1, +1\}$, our goal is to (i) produce a well-exposed image I_{en} and (ii) estimate dense optical flow $V \in \mathbb{R}^{H \times W \times 2}$ for the same interval.

Voxel representation. The raw event stream $\mathcal{E}$ is an irregular set of tuples (x_k, y_k, t_k, p_k) that is not directly compatible with dense CNN or transformer backbones. To obtain a

tensor of fixed dimension we partition the observation window $[t_0, t_0 + \Delta T]$ into B uniform sub-intervals of length $\Delta t = \Delta T/B$ and *accumulate* events that fall into the same spatio-temporal voxel. In this way, asynchronous event streams are discretized into voxel grids over a specific time window, capturing both polarities and short-term temporal evolution (Ding et al., 2022; Zhu et al., 2019). This voxel grid is then fed into the event feature encoder. Formally, the voxel grid $E \in \mathbb{R}^{H \times W \times B}$ is defined as

$$E(x, y, b) = \sum_{k: \lfloor \frac{t_k - t_0}{\Delta t} \rfloor = b} p_k \delta_{x, x_k} \delta_{y, y_k}, \quad b = 0, \dots, B-1,$$

where δ is the Kronecker delta. The grid therefore encodes polarity-weighted event counts at pixel (x, y) during temporal bin b , preserving local motion contrast while converting the asynchronous stream into a dense 3-D tensor amenable to standard convolutional or attention operations.

Network overview. As shown in Fig. 3, the ECDDP backbone f_E converts the event voxel grid into features $\mathbf{F}_E$. In parallel, a lightweight image encoder f_I processes a preliminarily light-up version of I to obtain $\mathbf{F}_I$. The two feature streams are fused, as shown in Fig. 4 and fed to a dual-branch decoder: one branch reconstructs the enhanced image I_{en} , whereas the second, coarse-to-fine pyramid branch predicts the dense optical-flow field V .

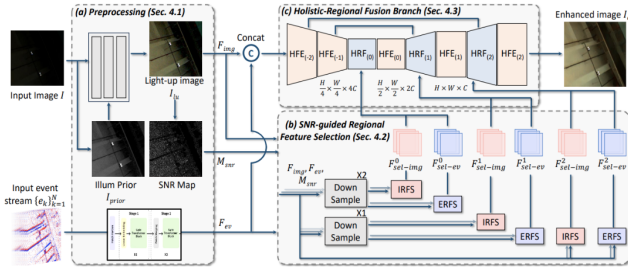


Figure 3. Overview of the modified *EvLight* pipeline. The original single-layer CNN-based event feature encoder is replaced by an ECDDP SWIN-T7 backbone that delivers rich event features, where the output of the second layer is forwarded. SNR-guidance injects these event features more aggressively where the RGB features become degraded.

Simply fusing event and RGB features can introduce artifacts in regions where events are sparse or noisy. To adaptively modulate the contribution of each modality, a *signal-to-noise ratio* (SNR) map is calculated. Regions with low-SNR receive event features more aggressively, while well-exposed RGB areas remain dominated by $\mathbf{F}_I$. This SNR-guided module allows the network to inject event cues *more aggressively* where the RGB features degrades, as illustrated qualitatively in Fig. 3.

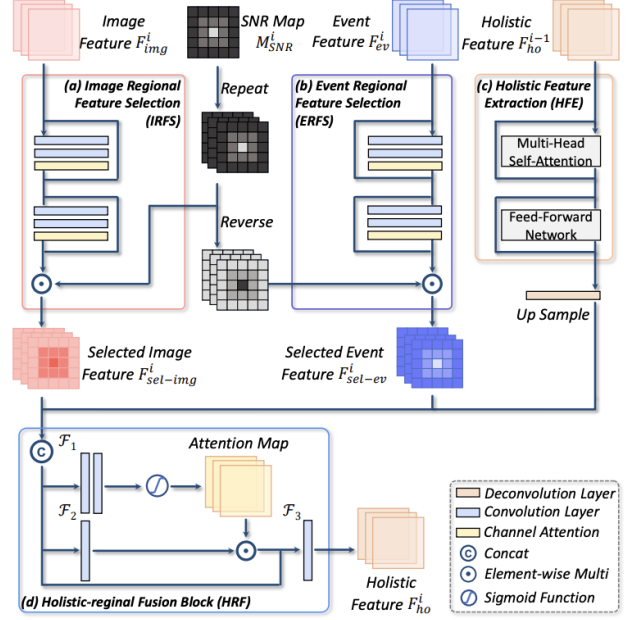


Figure 4. Details of each block in SNR-guided Regional Feature Selection and Holistic-Regional Fusion Branch's decoder.

Training objective. The network is optimised end-to-end by minimising the total loss:

$$\mathcal{L}_{\text{total}} = \lambda_{\text{enh}} \mathcal{L}_{\text{enh}} + \lambda_{\text{flow}} \mathcal{L}_{\text{flow}} + \lambda_{\text{consist}} \mathcal{L}_{\text{consist}}, \quad (1)$$

where λ_{enh} , λ_{flow} , and λ_{consist} balance the tasks, and:

- $\mathcal{L}_{\text{enh}}$ is the enhancement loss
- $\mathcal{L}_{\text{flow}}$ is the flow loss (optionally multi-scale)
- $\mathcal{L}_{\text{consist}}$ combines consistency terms:

$$\mathcal{L}_{\text{consist}} = \lambda_{\text{photo}} \mathcal{L}_{\text{photo}} + \lambda_{\text{warp}} \mathcal{L}_{\text{warp}} + \lambda_{\text{temp}} \mathcal{L}_{\text{temp}} \quad (2)$$

Enhancement Loss. The enhancement loss combines a Charbonnier-penalized intensity term with a perceptual distance metric computed in *AlexNet* feature space:

$$\mathcal{L}_{\text{enh}} = \sqrt{\|I_{\text{enh}} - I_{\text{gt}}\|_2^2 + \varepsilon^2} + \lambda \|\Phi(I_{\text{enh}}) - \Phi(I_{\text{gt}})\|_1, \quad (3)$$

where:

- $\varepsilon = 10^{-4}$ is the Charbonnier constant for numerical stability
- λ controls the weight of the perceptual term
- $\Phi(\cdot)$ denotes feature extraction using a frozen *AlexNet* backbone

- I_{enh} and I_{gt} represent the enhanced and ground truth images respectively

Flow Loss. The optical flow is supervised using a robust Charbonnier-penalized end-point error (EPE):

$$\mathcal{L}_{\text{flow}} = \frac{1}{|\Omega|} \sum_{x \in \Omega} \sqrt{(u_x - U_x)^2 + (v_x - V_x)^2 + \varepsilon^2}, \quad (4)$$

where:

- (u_x, v_x) and (U_x, V_x) represent the predicted and ground-truth flow vectors at pixel x , respectively
- Ω denotes the set of valid pixels
- $\varepsilon = 10^{-3}$ provides numerical stability and prevents division-by-zero

For multi-scale flow predictions, the loss is computed at each scale and aggregated with geometrically decaying weights:

$$\mathcal{L}_{\text{flow}}^{\text{multi-scale}} = \sum_{s=0}^{S-1} \gamma^s \mathcal{L}_{\text{flow}}^s, \quad (5)$$

where s indexes the scale level, S is the total number of scales, and $\gamma \in (0, 1)$ is the decay factor for coarser scales.

Consistency losses. In addition to the task-specific objectives, we impose three auxiliary consistency terms that stabilise training and improve temporal coherence.

1. **Photometric consistency** encourages the image obtained by warping the source frame with the predicted flow to match the target frame:

$$\mathcal{L}_{\text{photo}} = \frac{1}{|\Omega|} \sum_{x \in \Omega} \sqrt{\|I_{\text{warp}}(x) - I_{\text{tgt}}(x)\|_2^2 + \varepsilon^2}, \quad (6)$$

where I_{warp} is the source image spatially warped by the predicted flow, I_{tgt} is the target image, Ω is the set of valid (non-occluded) pixels, and ε is the Charbonnier constant that makes the penalty differentiable and robust to outliers.

2. **Flow-warping consistency** penalises disagreement between the forward flow $\mathbf{f}_t$ and the backward flow $\mathbf{f}_{t+1 \rightarrow t}$ (obtained by warping $\mathbf{f}_{t+1}$ with $\mathbf{f}_t$):

$$\mathcal{L}_{\text{warp}} = \frac{1}{|\Omega|} \sum_{x \in \Omega} \sqrt{\|\mathbf{f}_t(x) - \text{warp}(\mathbf{f}_{t+1}, \mathbf{f}_t)(x)\|_2^2 + \varepsilon^2}. \quad (7)$$

This term enforces geometric cycle consistency and suppresses spurious flow in occluded or low-texture areas.

3. **Temporal consistency** keeps the network’s consecutive predictions $\mathbf{P}_t$ and $\mathbf{P}_{t+1}$ (e.g. enhanced images) mutually coherent, except in regions that become occluded:

$$\mathcal{L}_{\text{temp}} = \sum_t \|\mathbf{M}_{t+1} \odot (\mathbf{I}_{\text{en}_t} - \mathbf{I}_{\text{en}_{t+1}})\|_1, \quad (8)$$

where $\mathbf{M}_{t+1}$ is a binary occlusion mask (1 if pixel remains visible, 0 otherwise) and $\odot$ denotes element-wise multiplication. This encourages smooth temporal transitions in the enhanced video stream.

4. Empirical Results

4.1. Datasets

SDE. SDE introduced by *EvLight* (Liang et al., 2024) contains 91 real sequences (43 indoor, 48 outdoor) captured with a DAVIS346 sensor with a resolution of 346x260. A robotic arm is used to repeat each camera trajectory twice—once in normal light and once behind an ND8 filter—to obtain ligned low/normal pairs.

SDSD. SDSD (*Seeing Dynamic Scene in the Dark*) (Wang et al., 2021) contains 125 full-HD video pairs (1920x1080) in which each scene is filmed twice: once under normal illumination and once behind an ND filter that emulates extreme darkness. A motorised pan-tilt platform ensures sub-pixel alignment between the paired sequences. Because an event camera was *not* available during acquisition, the authors of EVLIGHT augment SDSD with *synthetic* events. Each frame pair is first downsampled to 346 x 260—the native resolution of a DAVIS346 sensor—and then fed into the v2e simulator to generate polarity event streams.

DSEC. DSEC (*Driving Stereo Event Camera*) (Gehrig et al., 2021a) is a stereo camera dataset in driving scenarios that contains data from two monochrome event cameras and two global shutter color cameras in favorable and challenging illumination conditions. Image data is available in 8-bit PNG files at a resolution of 1440x1080 and is rectified. Event data is stored at VGA resolution (640x480).

4.2. Evaluation Metrics

We evaluate performance using both quantitative metrics and qualitative visualizations. For low-light image enhancement, we report fidelity metrics (PSNR, PSNR*) and perceptual quality (SSIM). For optical flow estimation, we measure geometric accuracy (EPE) and directional consistency (AE).

Peak Signal-to-Noise Ratio (PSNR): It measures pixel-level fidelity to ground truth:

$$\text{PSNR} = 10 \log_{10} \left(\frac{L^2}{\text{MSE}} \right); \text{MSE} = \frac{1}{MN} \sum_{i,j} (I_{ij} - \hat{I}_{ij})^2 \quad (9)$$

where L is the maximum pixel value (1 for normalized images, 255 for 8-bit), and MSE is the mean squared error.

PSNR* (PSNR-star): An improved variant that accounts for local luminance adaptation:

$$\text{PSNR}^* = 10 \log_{10} \left(\frac{L^2}{\text{MSE}^*} \right); \text{MSE}^* = \text{MSE} \cdot (1 + \alpha \text{TV}(\nabla I)) \quad (10)$$

where $\text{TV}(\nabla I)$ is the total variation of gradient magnitudes, and α weights the structural penalty (typically 0.1).

Structural Similarity Index (SSIM): It assesses perceptual quality through patch statistics:

$$\text{SSIM}(x, y) = \frac{(2\mu_x\mu_y + C_1)(2\sigma_{xy} + C_2)}{(\mu_x^2 + \mu_y^2 + C_1)(\sigma_x^2 + \sigma_y^2 + C_2)} \quad (11)$$

with μ, σ as local means/variances over a Gaussian window, and C_1 and C_2 to prevent instability.

Average Endpoint Error (EPE): It computes the Euclidean distance between predicted and ground truth flow vectors:

$$\text{EPE} = \frac{1}{|\Omega|} \sum_{(i,j) \in \Omega} \sqrt{(u_{x,ij} - \hat{u}_{x,ij})^2 + (u_{y,ij} - \hat{u}_{y,ij})^2} \quad (12)$$

where Ω excludes occluded/invalid pixels.

Average Angular Error (AE). It measures directional consistency in radians. Let $\mathbf{u}_{ij} = (u_{x,ij}, u_{y,ij}, 1)$ and $\hat{\mathbf{u}}_{ij} = (\hat{u}_{x,ij}, \hat{u}_{y,ij}, 1)$ be the ground-truth and predicted flow vectors augmented with a dummy depth component. AE is the mean 3-D angle between these vectors:

$$\text{AE} = \frac{1}{|\Omega|} \sum_{(i,j) \in \Omega} \cos^{-1} \left(\frac{\mathbf{u}_{ij} \cdot \hat{\mathbf{u}}_{ij}}{\|\mathbf{u}_{ij}\| \|\hat{\mathbf{u}}_{ij}\|} \right),$$

where Ω is the set of valid pixels. An AE of 0° denotes perfect directional agreement.

- All metrics are computed per-frame and averaged across the test set
- Flow metrics exclude invalid pixels
- SSIM uses the standard implementation from (Wang et al., 2004)

4.3. Preliminary Work

We first turned our attention to the ECDDP framework, and attempted to modify it for optical flow estimation. The ECDDP network consists of a SWIN-T7 based student-teacher network as shown in Figure 2 and has three similarity losses: patch-level similarity, context-level similarity and image-level similarity. We use the pre-trained network as provided by the authors and attempt to fine-tune it for dense

optical flow, since their reported evaluation on this dense prediction task appears to outperform every other method in the literature. We fine-tuned the SWIN-T7 based ECDDP backbone shown in more detail in Figure 5. We will only extract features from the second SWIN layer (comprising of a total of 4 transformer blocks) and use them as a replacement for the event feature extractor for EVLight as explained later on.

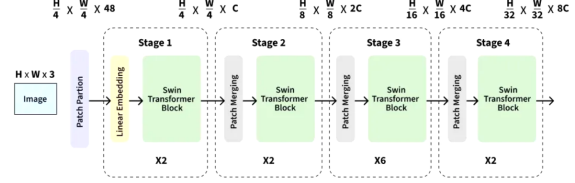


Figure 5. The SWIN-T7 backbone for the ECDDP network, we only concern ourselves with features from the second layer, which yields a $\frac{1}{8}$ down-sampling on the input resolution

The ECDDP framework consists of a momentum-updated teacher network supervising the student via three similarity losses: *patch-level similarity loss* to enforce local feature consistency, *context-level similarity* to mitigate the sparsity and background noise in event patches, and *image-level similarity* to ensure that global image semantics are aligned. All three objectives are equally weighted and combined into the final training loss:

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{patch}} + \mathcal{L}_{\text{context}} + \mathcal{L}_{\text{image}} \quad (13)$$

Hyperparameter	Value
Optimizer	AdamW
Peak learning rate	2×10^{-4}
Weight decay	10^{-4}
Betas (β_1, β_2)	(0.9, 0.999)
LR schedule	Warmup + Cosine decay
Warmup duration	1.5 epochs
Minimum learning rate	1×10^{-5}

Table 1. Fine-tuning settings

Table 2. Fine-tuned ECDDP performance on **DSEC-val**.

Sequence	EPE ↓
zurich_city_11.a	1.511
zurich_city_11.b	1.443
zurich_city_11.c	1.328
zurich_city_02.e	1.488
overall	1.420

Table 3. Average endpoint error (EPE) on **DSEC-val**.

Method	EPE ↓
E-RAFT	0.788
MultiCM	3.472
E-Flowformer	0.759
TMA	0.743
OF_EV_SNN	1.707
IDNet	0.719
ECDDP (ResNet50)	0.701
ECDDP (ViT-S/16)	0.714
ECDDP (Swin-T/7)	0.697
Ours (ECDDP fine-tuned)	1.420

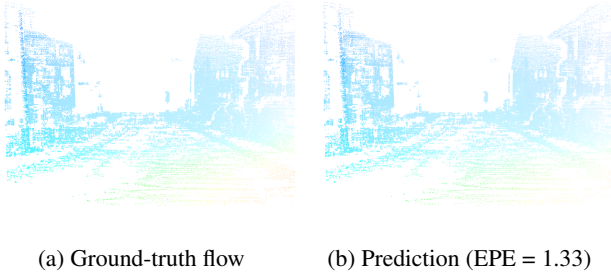


Figure 6. Qualitative comparison of optical-flow results after fine-tuning.

Remark. Despite faithfully following the authors’ published fine-tuning recipe, our replication of the ECDDP Swin-T/7 backbone does not reach the sub-0.70 EPE reported in the original paper. Nonetheless, the process gave us crucial implementation insight and validated key hyperparameters that we now transfer to our modified *EvLight* architecture, forming a solid background for the next round of experiments.

4.4. Pseudo-label Generation

While SDE provides ground-truth normal-light RGB frames, it lacks corresponding optical flow ground truth. This absence necessitates the generation of reliable dense optical flow pseudo-labels to supervise optical flow estimation. We employ state-of-the-art frame-based optical flow methods to generate pseudo-labels from consecutive ground-truth normal-light RGB frames:

$$\mathbf{F}_{t \rightarrow t+1} = \mathcal{M}(I_t, I_{t+1}) \quad (14)$$

where $\mathcal{M}$ represents our chosen flow estimation method and $\mathbf{F}_{t \rightarrow t+1}$ is the predicted flow field. We employ two the state-of-the-art models: **MemFlow** (Dong & Fu, 2024) with motion-enhanced memory networks and **DPFlow** (Morimitsu et al., 2025) using differentiable patch-match.

As shown in Fig. 7, our pseudo-label generation pipeline produces reliable optical flow for indoor sequences but fails

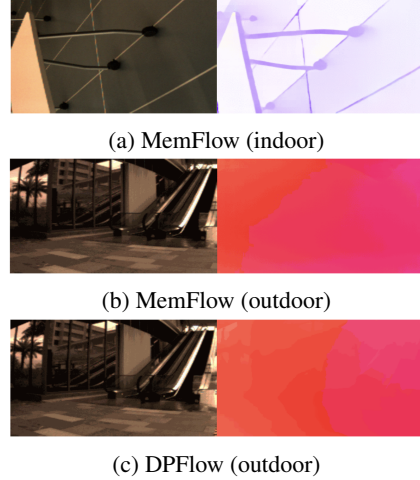


Figure 7. Pseudo optical flow label generation comparison showing MemFlow’s indoor success vs. outdoor failures, while DPFlow performs slightly better for outdoor sequences.


 Figure 8. DPFlow pseudo-label for an SDSD outdoor pair. Frame I_t (top-left), Frame I_{t+1} (top-right), and the predicted flow field $\mathbf{F}_{t \rightarrow t+1}$ (bottom).

for outdoor scenarios. We experimented with state-of-the-art flow-estimation models—most notably **FlowAnything** (Liang et al., 2025)—using mixed checkpoints. Despite these efforts, it still failed to produce reliable optical-flow maps for outdoor sequences.

SDSD outdoor pseudo-labels. We further assessed the SDSD outdoor sequences with DPFlow and found that it produces stable, high-quality flow fields. Because of this reliability, SDSD pseudo-labels will be leveraged in the subsequent training experiments. An illustrative example is shown in Fig. 8.

4.5. SDE-Indoor Training

We replace the original feature encoder with an **ECDDP** backbone and introduce a lightweight, multi-stage upsampling head to transfer the features from the second layer

of the ECDDP backbone that outputs 1/8th resolution of the original input size. We initialise the ECDDP backbone *from scratch* (no pre-trained weights).

The training recipe is summarised in Table 4. We run for 100 epochs with mixed-precision and a cosine LR schedule, validating every epoch. Losses combine a reconstruction term and a perceptual term with a 2:1 ratio, while evaluation relies on SSIM, PSNR, and PSNR*.

Parameter	Value
Training epochs	40
Batch size (train / val)	8 / 1
Optimiser	Adam (LR= 1.5×10^{-4} , <i>cosine</i>)
Mixed precision	✓
Base channels (C_0)	48
Voxel-grid channels	32
Loss weights	$\mathcal{L}_{\text{recon}} : 1.0, \mathcal{L}_{\text{perc}} : 0.5$
Metrics	SSIM, PSNR, PSNR*

Table 4. Training configuration

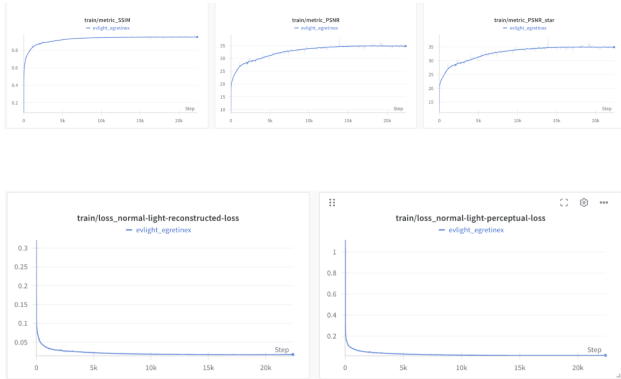


Figure 9. Training losses.

Table 5. Quantitative comparison on the indoor evaluation set. Higher is better for all metrics.

Method	SSIM $\uparrow$	PSNR $\uparrow$	PSNR* $\uparrow$
Ours	0.7407	21.34	24.40
EvLight	0.7697	22.44	24.81
(Liu et al., 2023)	0.7051	21.79	23.88
(Wang et al., 2020)	0.7277	21.25	23.19
(Jiang et al., 2023)	0.6168	19.98	21.44

4.6. Ablation Study

To disentangle the impact of (i) pre-training, (ii) the feature-map ‘‘tap’’ layer, and (iii) the *PixelShuffle*-upsampling de-

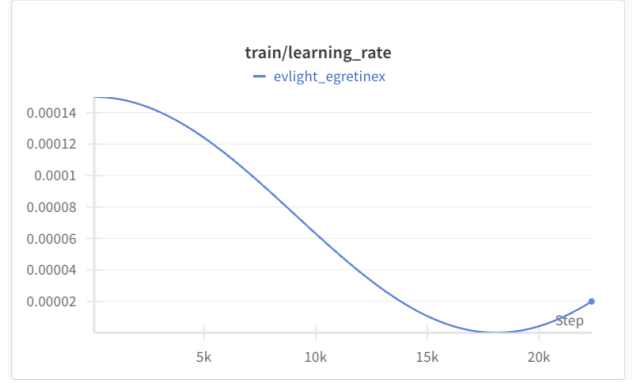


Figure 10. Learning-rate schedule employed during training.



Figure 11. Validation losses.

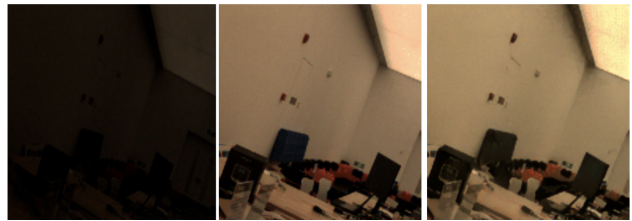


Figure 12. Qualitative comparison on an indoor low-light sample. From left to right: (a) input low-light image, (b) **EvLight** prediction, and (c) **ours**.

sign, we conduct six controlled experiments (Table 6). All settings follow the training recipe in Section 4.5.

Key observations. Overall, using pre-trained weights for the ECDDP backbone is not a decisive factor for indoor performance; tapping features from the 2nd encoder block yields a modest SSIM gain over the 1st-layer tap; and adding the smoother refinement to the PixelShuffle upsampler provides better features in general.

Table 6. Ablation results on the indoor evaluation set (higher is better).

Pre-train	Tap	Upsampling	SSIM	PSNR	PSNR*
–	2nd	PS×8	0.7012	20.78	23.67
✓	2nd	PS×8	0.7174	21.01	24.52
✓	1st	PS×4	0.7197	21.02	23.90
✓	2nd	PS×8 (smooth)	0.7370	20.97	23.94
✓	1st	PS×4 (smooth)	0.7201	21.24	23.86
–	1st	PS×4 (smooth)	0.7290	21.52	23.73

5. Future Work

Our baseline results confirm the potential of the proposed joint architecture, yet important challenges remain. We outline three immediate directions aimed at closing the remaining gaps.

- **Robust outdoor flow labels.** We will continue investigating classical and deep flow algorithms—e.g. cycle-consistent patch-match and memory-augmented networks—to derive reliable dense pseudo-labels for *SDE-outdoor*. Accurate motion supervision in these scenes should further stabilise the enhancement branch.
- **Dual-decoder training on SDS and SDE.** Leveraging the shared encoder’s success, we plan to introduce task-specific decoder heads and cross-task consistency losses. Experiments will begin on *SDS*, where DPFlow already provides reliable labels for both indoor and outdoor splits, and will later extend to *SDE-indoor*—and to *SDE-outdoor* once reliable pseudo-labels are available.
- **Ablations on SNR-guided event injection.** The SNR-guided injection module fuses event features at all scales. We will conduct controlled ablations that (i) disable the module entirely and (ii) vary its aggressiveness via a tunable gating factor. These studies will guide a better fusion strategy.

6. Conclusion

We have introduced a unified framework that simultaneously tackles *low-light image enhancement* and *event-based*

optical-flow estimation within a single network. Our architecture augments a motion-aware EvLight backbone with a Swin-T7 ECDDP encoder through bi-directional, cross-task consistency losses. In practice, motion cues from events guide the noisy, dark, and blurred RGB frames, while the enhanced RGB features in turn regularise optical flow prediction under extreme conditions.

Our experiments on the SDE-indoor confirm the potential of treating tightly coupled downstream tasks in a multi-task setting rather than in isolation.

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